

assets/cisuc_logo.png

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`assets/cisuc/logo.png`

- ▶ Doubts about novelty - Lack of Differentiation from prior work;
- ▶ Limited Theoretical Justification of parameters (Lack of Ablation Studies);
- ▶ Should Ablate the effect of each component of the proposed method;
- ▶ More datasets.

assets/mask_updates.png

A prediction is the sum of the products of the node decisions along the path to each leaf.

FADO: A Real-Time Online Product Recommendation Algorithm
Aware Online Learning for Product Recommendation

```
for each leaf  $\ell$ :  
  leaf_probs[ $\ell$ ]  $\leftarrow$  product(probs[:, :,  $\ell$ ])  
  # materialise every node-leaf pair  
  # repeat the product for every leaf  
prediction  $\leftarrow$  leaf_probs  $\times \theta$ 
```

reuse each partial path probability
prediction $\leftarrow$ leaf_probs $\times \theta$
Tree Depth

04/21

Complexity goes from $O(B \times 4^n)$ to $O(B \times 2^n)$, where n is the tree depth.

assets/probability_updater_performance.png

We can now ablate how trees’ depth affect the forest’s predictive and fairness metrics

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¹Stopped at n = 14 because the mask’s runtime became unbearable.

Complexity: $O(NW)$, where W is the window size.

counters.
Complexity: $O(NA)$, where A is the number of groups.

15.39× faster with identical DP and EO at every timestep. **06/21**

The counter-based window preserves the metric values while avoiding repeated scans of the same samples.

